

AccessibleHealthBench: Evaluating Large Language Model Adaptability to Financial, Cultural, and Lifestyle Health Constraints

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Abstract

Large language models (LLMs) are increasingly consulted for personal health advice on diet, nutrition, and fitness. However, these models may implicitly assume that users have unlimited grocery budgets, Western culinary backgrounds, and flexible schedules. We introduce **AccessibleHealthBench**, a reproducible benchmark of 120 prompts spanning financial, cultural, and lifestyle accessibility constraints. We evaluate multiple LLMs using structured extraction, similarity analysis, pairwise comparison, LLM-as-judge evaluation, and human validation. Our findings show that models often adapt superficially to user constraints while still producing advice that may not be practically accessible.

1 Introduction

Artificial intelligence is increasingly being used for health and nutrition guidance. Many users now consult large language models for meal plans, diet advice, workout recommendations, and wellness suggestions. Prior work shows that artificial intelligence can support personalized dietary recommendations and nutrition systems. Large language models such as ChatGPT, Claude, DeepSeek, and Llama can generate structured and fluent health recommendations. However, these systems may also implicitly assume ideal conditions about users, such as access to expensive foods, gym memberships, or flexible schedules.

For example, AI-generated meal plans may frequently recommend salmon, quinoa, or avocado without considering grocery budgets. Similarly, fitness plans may assume that users have significant free time or access to equipment. These assumptions can make otherwise accurate health advice unrealistic for many individuals.

This issue is important because health advice is only useful if users can practically follow it. Recommendations that ignore financial limitations, cul-

tural dietary traditions, or demanding lifestyles may unintentionally exclude many users.

Although previous studies show that LLMs perform strongly on nutrition knowledge evaluations, knowledge alone does not guarantee accessibility or inclusivity.

This project investigates whether LLM-generated nutrition and fitness recommendations effectively adapt to real-world user constraints. In particular, the study focuses on three key accessibility dimensions: financial accessibility, cultural dietary relevance, and lifestyle feasibility.

The societal implications of this problem are significant because AI-generated health recommendations may increasingly influence everyday lifestyle and wellness decisions. If these systems fail to account for financial, cultural, or lifestyle diversity, they may unintentionally reinforce existing health inequalities and reduce the practical usefulness of AI-driven guidance for underrepresented communities.

Additionally, accessibility-aware health recommendation systems have the potential to improve inclusivity and broaden access to personalized wellness support, particularly for individuals who may not have regular access to professional nutrition or fitness consultation services. Our main contributions are:

1. A benchmark dataset covering financial, cultural, and lifestyle accessibility constraints.
2. A multi-model evaluation pipeline using automated analysis, pairwise comparison, and human validation.
3. An investigation into whether LLMs meaningfully adapt health advice to realistic user constraints.
4. A discussion of societal and ethical implications of inaccessible AI-generated health recommendations.

1.1 Differences Between Proposal and Final Report

The final project differs from the original proposal in several important ways. Initially, the project primarily focused on automated evaluation approaches. However, during implementation, the evaluation framework was expanded to incorporate human validation and pairwise comparison methods, as absolute scoring alone was not sufficiently effective in capturing nuanced differences in response quality and accessibility adaptation.

We also refined the final model set based on practical considerations such as API accessibility, runtime constraints, and evaluation consistency across providers. In addition, the benchmark design evolved from standalone prompts into paired baseline and constraint-aware prompt structures, enabling more systematic analysis of model adaptation behavior under user-specific accessibility constraints.

2 Related Work

Artificial intelligence is increasingly used in nutrition research, personalized recommendation systems, and health-related applications. The references review artificial intelligence systems for personalized dietary recommendations and highlight the potential of AI-driven approaches for supporting individualized nutrition planning. Similarly, discuss the growing applications of artificial intelligence in personalized nutrition and food-related systems, emphasizing their role in improving dietary recommendation and health-support frameworks.

Other work examines AI-supported health infrastructure. It introduces AI4FoodDB, a framework that integrates nutrition data, wearable technologies, and AI-driven recommendation pipelines to support personalized health and lifestyle guidance. Broader review studies further indicate that artificial intelligence is playing an increasingly important role in nutrition-related decision support and health recommendation systems.

While these systems contribute significantly to personalization and scalability, most prior research primarily emphasizes recommendation accuracy and overall system performance, with comparatively limited focus on accessibility-related considerations.

Research on large language models has also identified broader concerns related to bias, representa-

tion, and societal risks. The studies specifically examine cultural bias and cross-cultural value differences in language models.

Recent LLM evaluation research has introduced pairwise comparison frameworks and LLM-as-judge methodologies for assessing response quality and model performance. Additionally, Sentence-BERT embeddings provide an effective approach for measuring semantic similarity between baseline and constraint-aware responses.

3 Data

3.1 Prompt Dataset

We created a benchmark dataset called **AccessibleHealthBench**. The dataset consists of 120 prompts, including 60 baseline prompts and 60 constraint-aware prompts. Each constrained prompt is paired with a corresponding baseline version, enabling systematic analysis of how model-generated recommendations adapt when user-specific accessibility constraints are introduced.

The prompts are divided into three categories: financial constraints, cultural dietary preferences, and lifestyle limitations.

Category	Baseline	Constrained	Total
Financial	20	20	40
Cultural	20	20	40
Lifestyle	20	20	40
Total	60	60	120

Table 1: AccessibleHealthBench dataset structure.

3.2 Model Responses

We evaluated responses generated by four large language models: GPT-4o-mini, Claude Haiku, DeepSeek Flash, and Llama 3.3 70B. Each model responded to all 120 benchmark prompts, resulting in a total of 480 generated responses for evaluation and analysis.

Model	Responses
GPT-4o-mini	120
Claude Haiku	120
DeepSeek Flash	120
Llama 3.3 70B	120
Total	480

Table 2: Total responses collected from all evaluated models.

4 Methods

Our evaluation framework integrates automated metrics, semantic similarity analysis, pairwise comparison, and human evaluation to provide a more comprehensive assessment of accessibility and response quality.

4.1 Structured Extraction

We extracted food entities, ingredients, activity routines, and constraint-related signals from model responses using NLP-based processing pipelines. This structured extraction process enabled more systematic and consistent analysis of long free-text responses across different evaluation dimensions.

4.2 Cost Estimation

To evaluate financial accessibility, extracted food items were mapped to external grocery pricing references and affordability indicators. This analysis was designed to examine whether responses that claimed to be budget-friendly actually included cost-conscious food choices and meaningful pricing-related guidance.

4.3 Cultural Diversity Analysis

To evaluate cultural bias, foods and activities were categorized based on their cuisine type and cultural context. A Western-centricity ratio was then computed to analyze the extent to which generated recommendations relied on Western dietary patterns and assumptions.

4.4 Constraint Sensitivity Analysis

We compared baseline and constrained responses using Sentence-BERT similarity analysis to measure the extent of semantic adaptation introduced after adding user-specific accessibility constraints. Higher similarity scores generally indicate limited response adaptation, whereas lower similarity scores suggest more substantial constraint-aware modifications in the generated recommendations.

4.5 LLM-as-Judge Evaluation

We used a structured rubric-based evaluation framework inspired by recent LLM assessment methodologies to systematically evaluate response quality across multiple accessibility dimensions. Responses were assessed for affordability, cultural relevance, feasibility, and overall accessibility consistency. Responses were systematically evaluated

across multiple accessibility dimensions, including affordability, cultural relevance, feasibility, and overall accessibility alignment.

4.6 Human Evaluation

Human evaluators independently reviewed a representative subset of responses and assigned scores on a scale of 1 to 5 across multiple evaluation dimensions, including affordability, cultural relevance, feasibility, and overall health quality. This evaluation helped assess how effectively the generated recommendations aligned with real-world accessibility and practical usability considerations.

5 Results

5.1 Financial Accessibility

The financial prompts evaluated whether models adapted recommendations to limited grocery budgets. The main result is that models often changed the ingredients in response to budget constraints, but they did not always provide specific prices or total cost estimates. This means that the advice appeared budget-conscious, but it was not always practically usable for a user with a strict grocery budget.

Provider	Abs. Score	Arena Win %	Cost Coverage
DeepSeek	4.97	78.9%	36.4%
Anthropic	4.83	63.7%	30.1%
OpenAI	4.80	32.2%	28.5%
Groq	4.88	24.7%	31.9%

Table 3: Financial accessibility results.

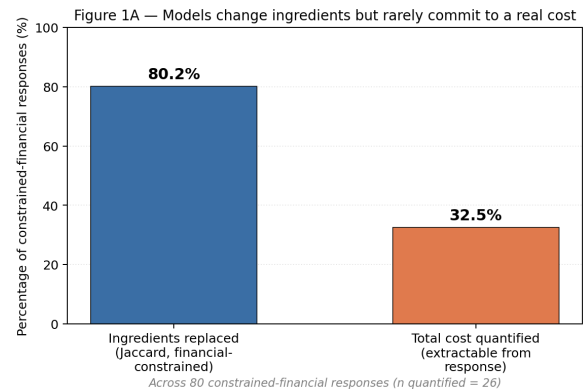


Figure 1: Ingredient replacement versus cost quantification.

The figure shows the difference between changing ingredients and actually pricing the recommendation. A model may replace expensive foods with

cheaper foods, but if it does not provide total cost information, the user cannot easily verify whether the plan fits their budget.

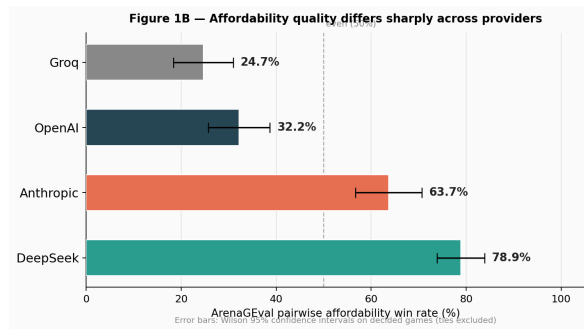


Figure 2: Affordability ranking across models.

The pairwise ranking shows that DeepSeek performed strongest on financial accessibility, followed by Anthropic. OpenAI and Groq performed lower in this dimension, especially when judged through pairwise comparison rather than only absolute scores.

5.2 Cultural Bias

The cultural prompts evaluated whether models adapted recommendations to non-Western dietary traditions. The results show that most models attempted to include culturally relevant foods when explicitly prompted, but the depth of adaptation varied. Some responses added cultural words or dishes without fully changing the structure of the recommendation.

Provider	Abs.	Arena %	W-Cent.	Adapt.
DeepSeek	5.00	83.3%	3.6%	0.282
Anthropic	4.99	36.7%	3.4%	0.369
OpenAI	4.96	50.0%	1.5%	0.270
Groq	4.97	29.8%	2.4%	0.281

Table 4: Cultural accessibility evaluation.

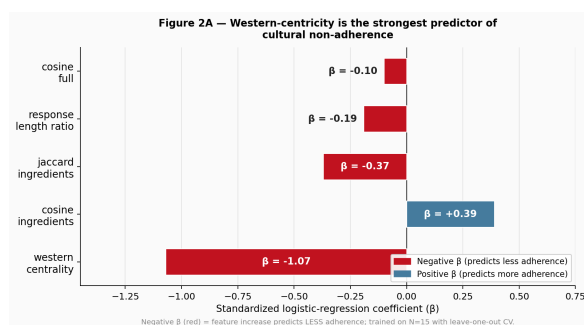
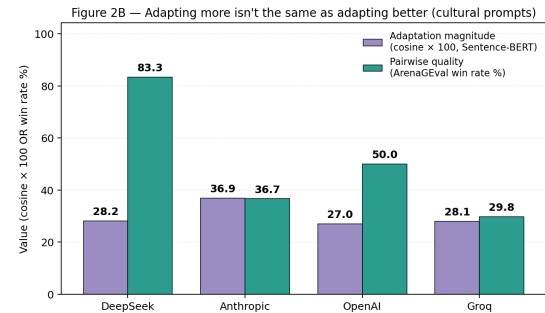


Figure 3: Western-centricity predictor analysis.

The Western-centricity analysis suggests that cultural accessibility is not only about mentioning one or two culturally specific foods. A stronger response should use culturally grounded ingredients, meal structures, and preparation styles throughout the recommendation.



Anthropic adapts MOST (left bars) but ranks 3rd in pairwise quality (right bars). DeepSeek wins despite less adaptation.

Figure 4: Adaptation magnitude versus quality.

This result shows that changing a response more does not always mean improving it. Some models produced larger differences from the baseline response, but those changes were not always judged as higher quality.

5.3 Lifestyle Constraints

The lifestyle prompts evaluated whether models generated realistic advice for users with time and schedule limitations. This dimension was more concrete than cultural relevance because time limits and equipment access can be checked more directly.

Provider	Abs. Score	Arena Win %	MET Coverage
DeepSeek	4.32	87.2%	7.3%
Anthropic	4.25	51.7%	18.4%
OpenAI	4.18	41.7%	10.8%
Groq	4.08	19.4%	17.3%

Table 5: Lifestyle feasibility evaluation.

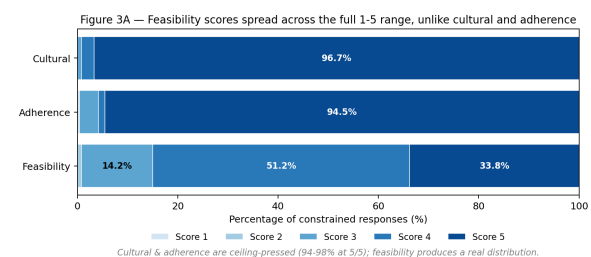


Figure 5: Score distributions across categories.

The score distribution shows that feasibility created clearer differences between model responses.

Advice that gives exact time blocks, simple routines, and realistic alternatives was judged more useful than vague recommendations.

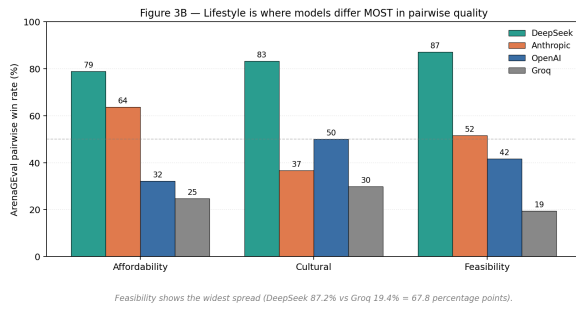


Figure 6: Pairwise comparison grouped by research question.

The pairwise comparison shows that DeepSeek had the strongest overall performance across the three research questions, while other models showed mixed performance depending on the constraint type.

5.4 Overall Results

Across all three dimensions, the overall results show that model performance is not uniform. A model that performs well on financial accessibility may not perform equally well on cultural adaptation or lifestyle feasibility.

Provider	Afford.	Cultural	Feasib.	Overall
DeepSeek	78.9%	83.3%	87.2%	83.1%
Anthropic	63.7%	36.7%	51.7%	50.7%
OpenAI	32.2%	50.0%	41.7%	41.3%
Groq	24.7%	29.8%	19.4%	24.6%

Table 6: Overall accessibility evaluation.

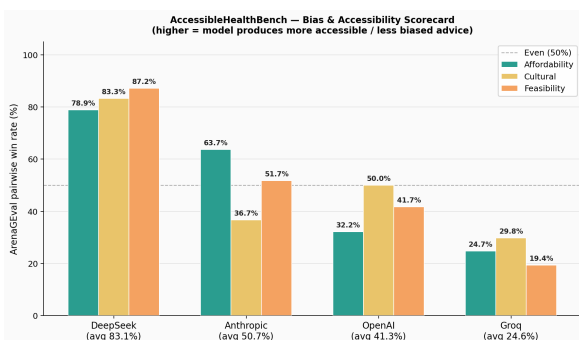


Figure 7: Overall accessibility scorecard.

The overall scorecard summarizes the main comparison across providers and constraint categories.

It shows that pairwise evaluation reveals differences that may be hidden when only absolute scores are used.

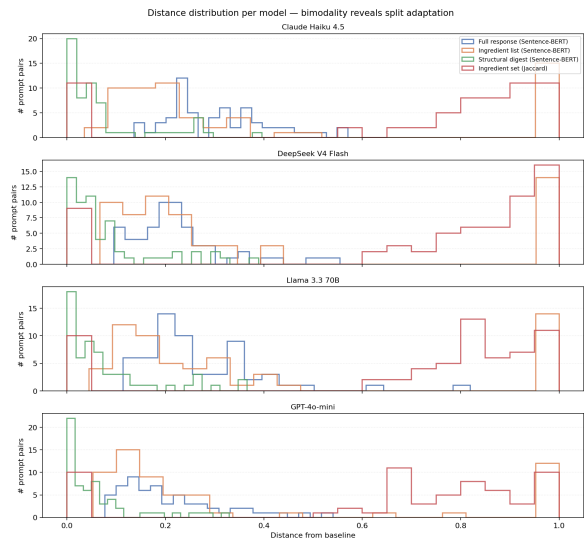


Figure 8: Distance distributions between baseline and constrained responses.

The distance distribution suggests that models often behave in two modes: either they barely change the baseline advice, or they rewrite most of the response. There are fewer cases where the model makes careful targeted substitutions.

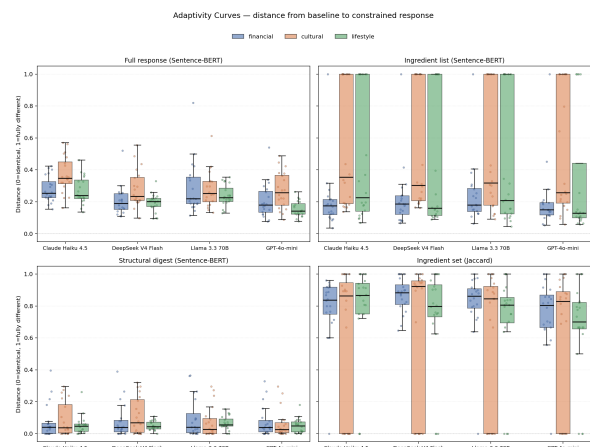


Figure 9: Adaptivity curves across constraint difficulty levels.

The adaptivity curves further show that adaptation is not always smooth or proportional to the difficulty of the constraint. Some models adapt strongly only after the prompt becomes very explicit.

5.5 Human Validation

Human validation was used to check whether the automated evaluation aligned with human judgment. Agreement was stronger for concrete dimensions such as affordability and feasibility, while cultural relevance was more subjective.

Dimension	Inter-human	LLM vs. Human	κ
Affordability	73%	80%	0.082
Cultural	40%	60%	0.102
Feasibility	87%	93%	0.100

Table 7: Human validation agreement.

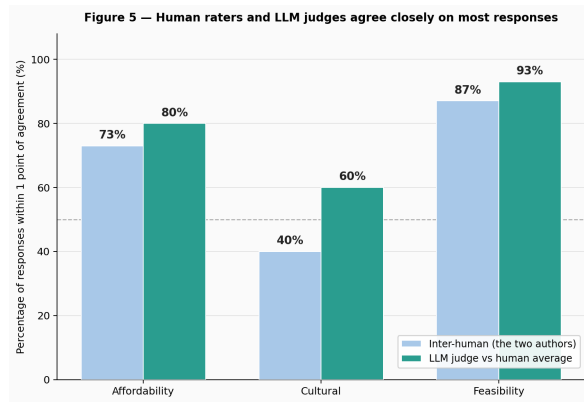


Figure 10: Human validation analysis.

The human validation results show that automated evaluation can be useful, but it should not be treated as a complete replacement for human judgment, especially when evaluating cultural appropriateness.

6 Discussion

Our results show that LLMs are generally capable of recognizing accessibility constraints, though the depth of adaptation varies across different scenarios. Financially constrained prompts often encouraged ingredient substitutions, but many responses provided limited explicit cost estimation. Cultural prompts demonstrated partial adaptation while still reflecting certain Western dietary defaults, and lifestyle prompts showed that generalized recommendations remain common even under strict time constraints.

These findings highlight the importance of developing future LLM health systems with stronger constraint-aware reasoning capabilities alongside fluent and informative response generation.

7 Ethical Considerations

This work highlights a few important ethical considerations. First, LLM-generated health advice may unintentionally reflect socioeconomic and cultural disparities present in training data. Second, automated evaluation methods may also introduce certain biases during assessment. Finally, AI-generated recommendations should be considered as supportive informational guidance and not as a replacement for professional medical expertise.

8 Limitations

This project presents several opportunities for future improvement. Human evaluation was conducted on a representative subset of responses to balance evaluation quality with practical time considerations. Expanding the scale of human evaluation in future work could further strengthen the reliability of the results.

Additionally, cultural accessibility is inherently context-dependent, and while the benchmark includes multiple cultural dietary scenarios, it cannot fully capture the diversity of global food traditions and regional practices. Future work may benefit from broader cultural coverage and more diverse evaluation settings.

Finally, the affordability analysis relied on external pricing references, where some region-specific ingredients had limited pricing coverage. Incorporating more comprehensive and region-aware pricing datasets could improve the precision of future evaluations.

9 Conclusion

This project evaluates whether large language models can generate health, nutrition, and fitness recommendations that remain accessible and practical under real-world financial, cultural, and lifestyle constraints. Through the AccessibleHealthBench benchmark, we systematically examined how modern LLMs adapt their responses when users specify limitations such as restricted grocery budgets, non-Western dietary traditions, demanding work schedules, or limited time availability.

Our findings indicate that current models are capable of recognizing and responding to accessibility-related prompts, but their adaptations are often surface-level rather than deeply constraint-aware. While many responses appear personalized and context-sensitive, they frequently fail to provide sufficiently practical, affordable, culturally

grounded, or realistically feasible guidance. For example, models may replace expensive ingredients with cheaper alternatives without offering concrete pricing information, mention culturally relevant foods without fully adapting the overall meal structure, or suggest lifestyle changes that remain unrealistic for users with severe time constraints.

The results further demonstrate that strong general nutrition knowledge does not necessarily translate into accessible or inclusive recommendations. Although several models achieved high absolute evaluation scores, pairwise comparison and human validation revealed meaningful differences in the practical usefulness and adaptability of their responses. Overall, this work highlights the importance of moving beyond fluency and correctness when evaluating AI-generated health advice and emphasizes the need for future systems that can perform more robust, context-aware, and human-centered accessibility reasoning.

10 Future Work

There are several opportunities to extend and improve this work in future research. First, the pairwise evaluation framework in this project relied primarily on ChatGPT-based judging. While this provided consistent evaluation, future work could incorporate multiple evaluator models such as Claude, Gemini, and Llama-based judges. Using a diverse ensemble of evaluators may reduce single-model evaluation bias and improve the robustness, reliability, and overall effectiveness of pairwise assessment results.

Second, the benchmark itself can be expanded to include a broader range of dietary and physical health scenarios. The current benchmark focuses mainly on nutrition accessibility, cultural adaptation, and lifestyle feasibility. Future versions could include additional dietary conditions, chronic disease considerations, mental wellness recommendations, fitness progression plans, and region-specific healthcare constraints. Incorporating a larger and more diverse set of physical health and dietary benchmarks would improve the generalizability of the evaluation framework.

Finally, the human evaluation dataset can be significantly expanded. The current human validation phase was conducted on a relatively small subset of responses due to time and resource limitations. Future work should include a larger and more diverse evaluator pool with broader demographic and

cultural representation. Increasing the size and diversity of the human evaluation dataset would improve the reliability of agreement analysis and provide a more comprehensive understanding of how different users perceive accessibility and practical usefulness in AI-generated health advice.

Acknowledgments

We thank our professor Dr Kai-Cheng Yang and Binghamton University for guidance and support throughout this project.

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